

CHEMISTRY X

Questions Distribution

According to the Chapters.

**This Distribution is carried out According to
NEW BOOK OF CHEMISTRY.**

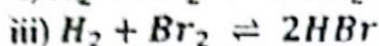
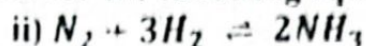
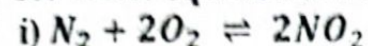
COMPILED BY M. Zubair Arif Yousufi

QUESTIONS		Years
UNIT:1 CHEMICAL EQUILIBRIUM		
Section "B"		
3. The value of K_c for the reaction is 1×10^{-4} $2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$ At a given temperature, the molar concentration of reaction mixture is, $HI = 2 \times 10^{-5} \text{ mol.dm}^{-3}$, $H_2 = 1 \times 10^{-5} \text{ mol.dm}^{-3}$ and $I_2 = 1 \times 10^{-5} \text{ mol.dm}^{-3}$. Predict the direction of the reaction. 12. Write three characteristics of equilibrium constant K_c .		2023
Section "C"		
14. State the law of mass action, apply it on the following reversible reaction. $aA + bB \rightleftharpoons cC + dD$ and derive the expression for equilibrium constant.		2024
SOME IMPORTANT QUESTIONS FROM NEW BOOK		
Section "B"		
1. Define chemical equilibrium with example. 2. Why chemical equilibrium is dynamic? 3. When writing an equation, how is a reversible reaction distinguish from irreversible reaction? 4. Write an equilibrium equation of monoatomic carbon and a molecule of oxygen as reactant and carbon monoxide as product. 5. Outline the characteristics of reversible reaction. 6. Distinguish between reversible and irreversible reaction. 7. State law of mass action. How is the active mass is represented? 8. Why equilibrium constant may or may not have unit? Justify with		

example.

9. How direction of a reaction can be predicted if K_c is known to you.

10. Write equilibrium constant expression for the following equations:



Section "C"

1. Describe dynamic equilibrium with two examples.

2. State law of mass action. Derive an expression for equilibrium

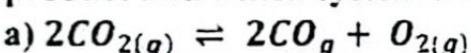
constant. 3. Describe the characteristics of equilibrium constant.

4. How can you predict the following stages of a reaction by comparing the values of K_c and Q_c .

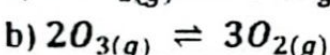
i. Net reaction proceeds in forward direction.

ii. Net reaction proceeds in reverse direction

5. Predict which system at equilibrium will contain maximum amount of product and which system will contain maximum amount of reactant?



$$K_c(927^\circ C) = 3.1 \times 10^{18}$$



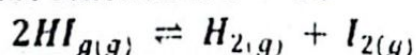
$$K_c(298K) = 5.9 \times 10^{55}$$

SOME IMPORTANT NUMERICALS FROM NEW BOOK

1. Dinitrogen tetra oxide N_2O_4 decomposed into nitrogen dioxide NO_2 in a reversible reaction. Derive equilibrium constant expression for the reaction of decomposition. Also interpret unit of K_c for balanced chemical reversible reaction.

2. PCl_5 , PCl_3 and Cl_2 are at equilibrium at 500K in a closed container and their concentrations are $0.8 \times 10^3 \text{ mol dm}^{-3}$, $1.2 \times 10^{-3} \text{ mol dm}^{-3}$ and 1.2×10^{-3} respectively. Calculate the value of K_c for the reaction along with unit.

3. The value of K_c for the reaction is 1×10^{-4}



At a given temperature, the molar concentration of reaction mixture is $HI = 2 \times 10^{-5} \text{ mol dm}^{-3}$, $H_2 = 1 \times 10^{-5} \text{ mol dm}^{-3}$ and $I_2 = 1 \times 10^{-5} \text{ mol dm}^{-3}$. Predict the direction of the reaction.

UNIT:2

ACID, BASE AND SALT

Section "B"

11. Find pH , pOH , $[OH^-]$ and $[H^+]$ of $2.46 \times 10^{-9} M$ KOH solution.

12. Write one use of each of the following salts:

(i) Hydrated calcium sulphate

(ii) Iron (II) sulphate hepta-hydrate

(iii) Potassium permanganate.

2025

3. Describe briefly two types of Buffer Solutions with examples.	2024
4. Elaborate the Arrhenius concept of acid and base with suitable examples.	
12. A solution of HCl has pH of 2.3 Calculate its pOH and $[H^+]$.	
4. What are the limitations of Arrhenius's Theory.	2023
5. Write three uses of salts.	
6. Find pH, pOH, $[OH^-]$ and $[H^+]$ of $2.46 \times 10^{-9} M KOH$ solution.	

Section "C"

14. State three different theories about concepts of acid and base, with example also indicate one limitation of each concept. Also complete conjugated acid and conjugated base of the following reactions:	2025																												
<table border="0"> <thead> <tr> <th>Acid</th> <th>+</th> <th>Base</th> <th></th> <th>Conjugated Acid</th> <th>+</th> <th>Conjugating Base</th> </tr> </thead> <tbody> <tr> <td>H_2SO_4</td> <td>+</td> <td>H_2O</td> <td>$\rightleftharpoons$</td> <td>_____</td> <td>+</td> <td>_____</td> </tr> <tr> <td>HCl</td> <td>+</td> <td>NH_3</td> <td>$\rightleftharpoons$</td> <td>_____</td> <td>+</td> <td>_____</td> </tr> <tr> <td>$\cdot CH_3COOH$</td> <td>+</td> <td>H_2O</td> <td>$\rightleftharpoons$</td> <td>_____</td> <td>+</td> <td>_____</td> </tr> </tbody> </table>	Acid	+	Base		Conjugated Acid	+	Conjugating Base	H_2SO_4	+	H_2O	$\rightleftharpoons$	_____	+	_____	HCl	+	NH_3	$\rightleftharpoons$	_____	+	_____	$\cdot CH_3COOH$	+	H_2O	$\rightleftharpoons$	_____	+	_____	
Acid	+	Base		Conjugated Acid	+	Conjugating Base																							
H_2SO_4	+	H_2O	$\rightleftharpoons$	_____	+	_____																							
HCl	+	NH_3	$\rightleftharpoons$	_____	+	_____																							
$\cdot CH_3COOH$	+	H_2O	$\rightleftharpoons$	_____	+	_____																							
17. What are salts? Write down its preparation by the action of acid on: with representative balanced chemical equation for each:																													
(i) Metal hydroxide (ii) Metal carbonate (iii) Metal oxide																													
16. What are salts? Write down its preparation by action of acid on:	2024																												
(i) metal hydroxide (ii) metal carbonate																													
(iii) metal oxide with representative chemical equation for each.																													
18. Define buffers. What is its composition? How are these utilized?	2023																												

SOME IMPORTANT QUESTIONS FROM NEW BOOK

Section "B"

- Discuss the properties of acid and base.
 - Elaborate the Arrhenius concept of acid and base with suitable example.
 - What is Bronsted-lowery acid-base theory?
 - What are conjugate acid base pairs? Explain with examples.
 - Classify the following solutions as acidic, basic or neutral.
- A solution that has $[H^+] = 1 \times 10^{-4} \text{ mol. dm}^{-3}$
 A solution that has $[H^+] = 1 \times 10^{-11} \text{ mol. dm}^{-3}$
 A solution that has $[OH^-] = 1 \times 10^{-9} \text{ mol. dm}^{-3}$
 A solution that has $[OH^-] = 1 \times 10^{-3} \text{ mol. dm}^{-3}$

6. Elaborate the ionization equation of water.
7. Define the following terms;
(a) pH (b) Indicator (c) Neutralization (d) Titration
8. Define buffers. What is the composition of buffers? Discuss its importance in our daily life.

Section "C"

1. Describe salts, preparation of salts and types of salts.
2. Explain in detail that how water ionization is related with pH of solution?
3. Discuss in detail how different solutions in aqueous system exhibit increase in acidity and increase in basicity on the basis of H^+ and OH^- ?
4. What do you mean by balancing of neutralization reaction with the help of examples?
5. Write down the uses of salt in daily life.

SOME IMPORTANT NUMERICALS FROM NEW BOOK

1. Calculate pH of 5M solution of NaOH
2. A solution of H_2SO_4 has pH of 1.05 calculate its pOH and $[H^+]$
3. The hydrogen ion concentration of a solution is $1 \times 10^{-6} \text{ mol. dm}^{-3}$ what is pH of the solution?

UNIT:3

ORGANIC CHEMISTRY

Section "B"

- | | |
|---|-------------|
| 2. Define Organic Chemistry and write two general characteristics of organic compounds. | 2025 |
|---|-------------|

5. Frame structural formulae of the following:

(i) 2,3 dimethyl pentane (ii) 3-ethyl,1-hexene (iii) pentyne

- | | |
|---|-------------|
| 5. Frame structural formula of the following Hydro-Carbons: | 2024 |
|---|-------------|

(i) 2,3 - dimethyl pentane (ii) 3 - ethyl, 1 - hexene
(iii) 2 - butyne

7. State vital force theory and how was this theory discarded by Friedrich Wohler?

7. What do you mean by Vital Force Theory?

8. Write down the molecular, structural and condensed formulae of the following compounds.

Compound	Molecular formula	Structural formula	Condensed formula
(i) Methane			
(ii) Ethane			
(iii) Propane			

3. Write down any three uses of 'Ethylene'.

2022

7. Complete and balance the following chemical equations of combustion reaction: (i) $\text{CH}_4 + \text{O}_2 \longrightarrow$ (ii) $\text{C}_2\text{H}_4 + \text{O}_2 \longrightarrow$ (iii) $\text{C}_2\text{H}_2 + \text{O}_2 \longrightarrow$	
13. Define the terms: (i) Homologous series (ii) Isomerism (iii) Functional group (iv) Aromatic Compounds	2018
17. Define: (i) Homologous Series (ii) Functional group (iii) Isomerism 20. A hydrocarbon contains six carbon atoms. Write its molecular formula. If it is: (i) An alkane (ii) An Alkene (iii) An alkyne	2017
20. Define: (i) homologous series (ii) Cracking (iii) Isomerism	2016
6. Write the names of functional groups and formulae of the following: (i) Ethyl Alcohol (ii) Acetic Acid (iii) Propyl Iodide 17. Define refining of Petroleum and write two fractions of it. 20. Define: (i) Functional Group (ii) Homologous series (iii) Isomerism	2015
Section "C"	
17. Differentiate between saturated and unsaturated hydrocarbons.	2023
17. (b) What are hydrocarbons? How these are classified? Explain in detail. Also draw structural diagrams.	2019
23(b). Define: (i) Hydro Carbons (ii) Aliphatic Hydro Carbons (iii) Alicyclic Hydro Carbons (iv) Aromatic Hydro Carbons.	2015
SOME IMPORTANT QUESTIONS FROM NEW BOOK	
Section "B"	
1. Define Vital Force Theory. 2. Explain how petroleum is source of organic compounds? 3. Define the functional group. Write the functional groups which contain carbon, hydrogen and oxygen. 4. Define the alkyl radicals with suitable examples. 5. What is homologous series? Name the some common homologous series 6. Identify the functional groups in the following compounds (a) $\text{CH}_3 - \text{CHO}$ (b) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{OH}$ (c) $\text{CH}_3 - \text{CO} - \text{CH}_3$ (d) $\text{CH}_3 - \text{COOH}$ (e) $\text{CH}_2 = \text{CH} - \text{CH}_3$ 7. Write the condensed structural formulas of the pentane and octane.	

8. What is catenation? Give any two examples of catenation of carbon atoms.

Section "C"

1. Give the important characteristics of organic compounds.
2. Differentiate between saturated and unsaturated hydrocarbons.
3. What are the main sources of organic compound? Mention with special reference of coal, petroleum and natural gas.
4. Describe the uses of organic compound.
5. Name the alkenes and alkynes having the following formula.
 (i) C_2H_4 (ii) C_3H_4 (iii) C_3H_6 (iv) C_6H_{12}
 (v) C_2H_6 (vi) C_8H_{16} (vii) C_7H_{12} (viii) C_6H_{10}
6. Define nomenclature and describe the I.U.P.A.C nomenclature rules for alkynes.
7. What do you mean by diversity and magnitude of organic compounds?

UNIT:4

BIOCHEMISTRY

Section "B"

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|---|------|
| 4. What are carbohydrates? Differentiate between aldoses and ketoses | 2025 |
| 9. Write one use of the following Enzyme:
(i) Yeast (ii) Cellulose (iii) Amyloze | 2024 |
| 10. How vegetable oil is converted into standard fat (ghee)? | |
| 9. Define the following:
(i) Aldose (ii) Ketose (iii) Saccharides | 2023 |
| 10. Elaborate Amino acids as monomers or building block of protein. | |

Section "C"

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|--|------|
| 17. What are amino acids? Describe the formation of dipeptide through peptide linkage bond between two amino acid units with structural equation. | 2024 |
| 18. What are carbohydrates? Classify carbohydrates on the basis of hydrolysis into three types with an example of each also mention the source from where these can be obtained. | |
| 16. Distinguish between fat soluble and water soluble vitamin. | 2023 |

SOME IMPORTANT QUESTIONS FROM NEW BOOK

Section "B"

1. What are the proteins?
2. Define the importance of deoxyribose nucleic acid (DNA).

3. Differentiate between fats and oils.
4. What are the polysaccharides? How monosaccharides are produced from polysaccharides?
5. What is peptide bond? How it is formed? And also explain dipeptides and tripeptides.
6. Enumerate the important uses of lipids.
7. What are the amino acids and give their general structure?
8. What is vitamin D? Give its sources and importance.
9. Distinguish between fat and water soluble vitamins.

Section "C"

1. What are the carbohydrates? Explain sources and uses of carbohydrates.
2. What are lipids? Write down the sources and uses of lipids.
3. Describe vitamins and types of vitamins.
4. Describe in detail nucleic acids, RNA and DNA.
5. How you can justify that deficiency of different types of vitamins causes dam' in human beings.

UNIT:5 ENVIRONMENTAL CHEMISTRY-I (THE ATMOSPHERE)

Section "B"

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| 7. What are the three spreading factors of pollution? Describe them. | 2025 |
| 8. What is meant by acid rain and how it demolishes monuments? | |
| 6. What is the atmosphere? List its five major layers. | 2024 |
| 11. Write down any three adverse impact of Global Warming. | |

Section "C"

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| 15. Elaborate Greenhouse effect (Global Warming) | 2025 |
| 19. Describe how does Ozone depletion occur in stratosphere by chlorofluoro carbons (CFCs) also indicate any three harmful impacts of Ozone depletion. | 2024 |
| 19. Describe Global warming. | 2023 |

SOME IMPORTANT QUESTIONS FROM NEW BOOK

Section "B"

1. Enlist major air pollutants and their sources.
2. Describe the effects of some air pollutants on human health?
3. What is the cause of acid rain?

4. Justify that Greenhouse effect leads to global warming.
5. List down the layers of atmosphere.
6. Write down the effects of acid rain.
7. Justify that change in altitude change the temperature of atmosphere
8. What are primary and secondary air pollutants.

Section "C"

1. Define atmosphere and explain its composition
2. Differentiate between stratosphere and troposphere.
3. Describe that how different air pollutants effects environment and human health.

UNIT:6 ENVIRONMENTAL CHEMISTRY-II (WATER)

Section "B"

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| 3. Write down three physical properties of water. | 2025 |
| 11. How can we remove temporary hardness of water by Clark's method. | 2023 |

Section "C"

- | | |
|--|-------------|
| 16. Describe any six water borne diseases. | 2025 |
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SOME IMPORTANT QUESTIONS FROM NEW BOOK

Section "B"

1. Describe composition of water.
2. Define hard and soft water
3. Describe water pollutant.
4. How we can remove temporary hardness of water?
5. List down the diseases due to polluted drinking water.
6. Differentiate between hard and soft water.

Section "C"

1. Write down the methods for removal of permanent hardness of water.
2. Describe the water pollutants in industries.
3. Justify that "Water is solvent".
4. Describe disadvantages of hard water?
5. Explain in detail Water borne diseases.

UNIT:7 ANALYTICAL CHEMISTRY

Section "B"

6. Define	(i) Error	(ii) Precision	(iii) Accuracy	2025
8. Define:	(i) Titration	(ii) Titrant	(iii) Analyte	2024

Section "C"

15. What do you mean by quantitative analysis? Also define the following terms: (i) error (ii) systematic error (iii) random error (iv) accuracy (v) precision	2024
15. What is high performance liquid chromatography (HPLC)? Explain it.	2023

SOME IMPORTANT QUESTIONS FROM NEW BOOK**Section "B"**

1. What do you think which methods are faster classical or instrumental?
2. How will you compare the analytical techniques to one another?
3. Can you give some examples of error related to your life?
4. What do you mean by quantitative Analysis?
5. Why we use Potentiometric Analysis in advance instrumental methods?
6. How scientists are using Infrared spectroscopy in quality control of different industries?
7. List down the applications of Conductometry?

Section "C"

1. Distinguish between following:
(a) Quantitative Analysis and Qualitative Analysis
(b) Titrimetric Analysis and Gravimetric Analysis
2. Prove that instrumental analytical methods are more effective than classical analytical methods?
3. Describe Gas Chromatography in detail?
4. Justify that electrochemical methods depend upon Electrochemical Cells?

UNIT:8**INDUSTRIAL CHEMISTRY****Section "B"**

- | | |
|--|-------------|
| 9. What materials are required for preparation of soft drink? Name them. | 2025 |
| 10. Draw flowsheet of soap preparation. | |
| 2. Write a few points about society, technology & science. | 2023 |

Section "C"

18. Define Petroleum. Describe fractional distillation of it. Write name of any five fractions distilled from petroleum with hydrocarbon and show it by the help of labelled diagram.	2025
19. What is meant by origin of pharma? Elaborate any five beneficial impacts of pharmaceutical industry in our society.	
14. Explain the process of sugar preparation from sugarcane	2023

SOME IMPORTANT QUESTIONS FROM NEW BOOK**Section "B"**

1. Define saponification process.
2. Describe that NaOH or KOH are used in preparation of soap
3. List down the raw material needed for sugar preparation
4. Explain components of soft drinks
5. Define petroleum
6. Justify the petroleum is "Black Gold"

Section "C"

1. Describe fractions of petroleum in detail.
2. Write down the importance of pharma industry.
3. Describe the sugar preparation with the help of flow sheet diagram.
4. Draw stepwise preparation of soft drinks in flow sheet diagram.

REASON	Chapter	Years
13. Give Reason: (i) Why number of carbon compounds are more than of silicon compound?	3	2025
(ii) The higher the altitude of stratosphere the temperature is increased.	5	
13. Justify. (i) Why the coolness on top of a mountain is greater than at sea level?	5	2024
(ii) Water is considered a universal solvent. Why?	6	
13. Give reason of the following: (i) Petroleum is "Black Gold"	8	2023
(ii) Alkanes said to be saturated Hydrocarbons.	3	
DEFINITIONS	Chapter	Years
2. Define: (i) Organic Chemistry	3	2024
(ii) Bio Chemistry	4	
(iii) Environmental Chemistry	5	